

3.1 General properties of waves

5054 Paper 2 only, 2015 onwards. Year by year, with figures and mark schemes.

2015

May/June 5054/21

Question 4

- 4 Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

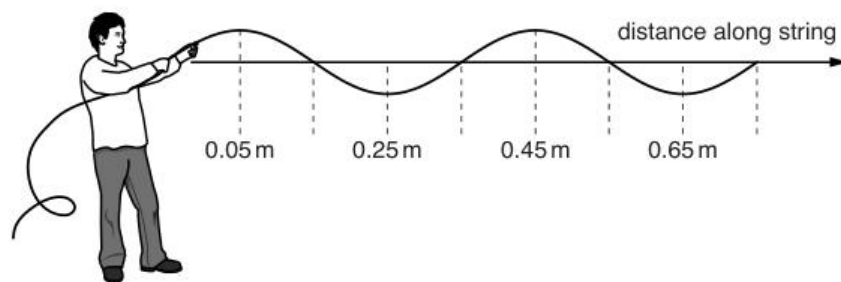


Fig. 4.1

- (a) Describe the movement of the particles of the string.

.....

[2]

- (b) Determine the wavelength of the wave.

wavelength =[1]

- (c) The speed of the wave is 2.0 m/s.

- (i) Calculate the frequency of the wave.

frequency =[2]

- (ii) On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

Question 9

- 9 Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

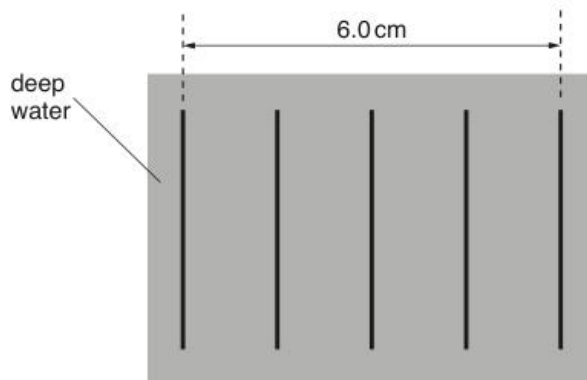


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

- (a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

- (b) (i) Determine the wavelength of the wave in deep water.

..... [1]

- (ii) Calculate the speed of the wave in deep water.

speed = [2]

- (c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

- (i) State and explain how this affects the wavelength of the wave.

.....

 [2]

- (ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

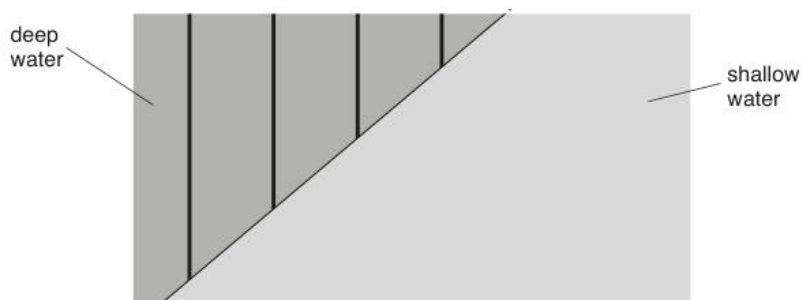


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water.

[3]

- (d) Sound is also a wave.

- (i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

- (ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

.....

.....

..... [3]

Question 7

7 Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

(a) Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

(b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

(i) State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

(ii) Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

(c) Describe one use of ultrasound waves.

.....
.....
.....
.....
..... [2]

Question 5

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

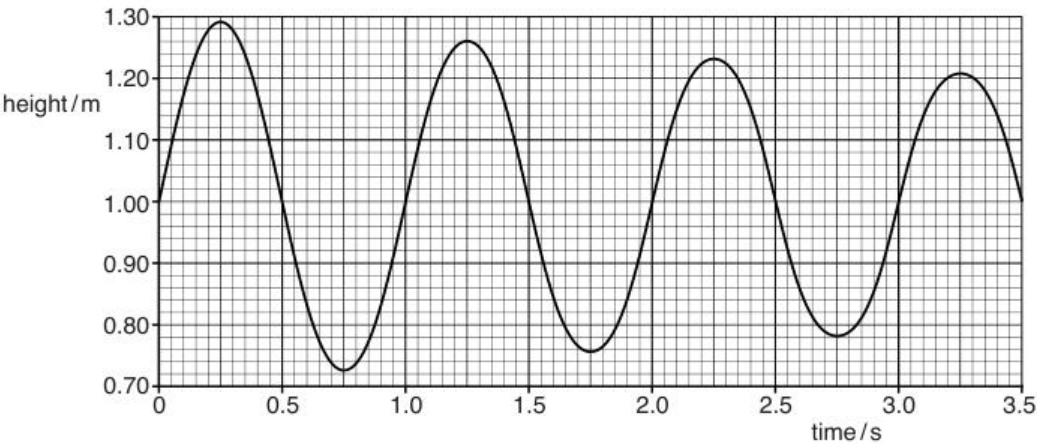


Fig. 5.2

- (i) Determine the amplitude of the vibration at time = 0.25s.
.....[1]
- (ii) Describe how the graph shows that the frequency of the wave is constant.
.....
.....[1]

Question 5

- 5 Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

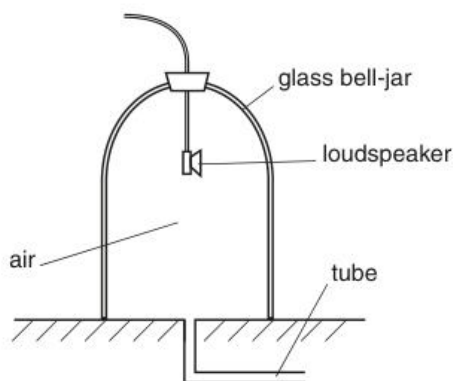


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

- (a) State what is meant by the *frequency* of a sound.

.....

 [1]

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength = [2]

- (ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....

[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....

[1]

2. how sound is transmitted through the air.

.....

[2]

- (ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....

[1]

Mark scheme — See official MS

2017

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Question 10

14

- 10 (a) Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

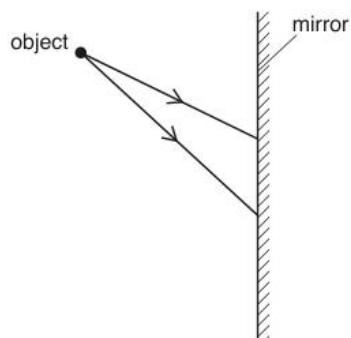


Fig. 10.1

- (i) On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

- (ii) Apart from its position, state one characteristic of the image.

.....[1]

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

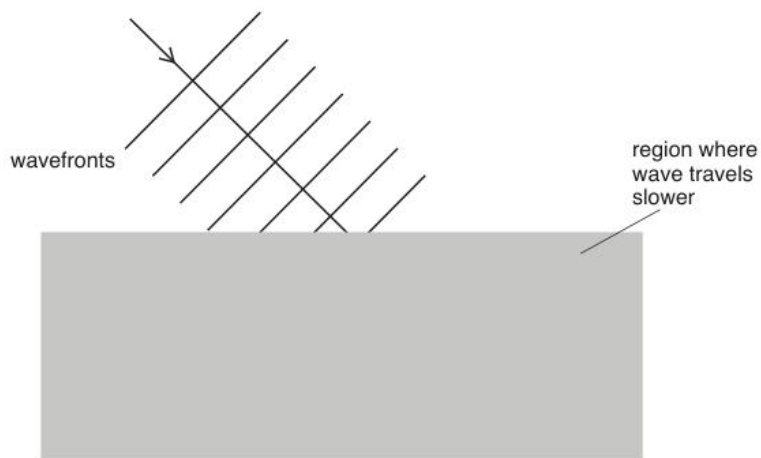


Fig. 10.2

- (i) State what is meant by *wavefront*.

.....
[2]

- (ii) Both reflection and refraction occur at the boundary.
1. On Fig. 10.2, draw the reflected wavefronts. [1]
 2. On Fig. 10.2, draw the refracted wavefronts. [2]
- (c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.
- (i) Calculate the speed of sound in air.

speed = [2]

- (ii) 1. State the range of frequencies that can be heard by a healthy human ear.
..... [1]
2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = [1]

- (iii) Describe a simple experiment to show that sound waves obey the law of reflection.
You may draw a diagram if you wish.

.....

 [3]

Mark scheme — 10(a)(i) both reflected rays correct by eye B1 image in correct position shown by continuation of rays behind mirror B1
 10(a)(ii) virtual or upright or laterally inverted or same size (as object) B1 10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1 10(b)(ii)1 correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye) B1 10(b)(ii)2 at least two lines at smaller angle to surface with correct orientation in slower medium B1 showing smaller and constant wavelength with wavefronts deviated in correct direction B1 10(c)(i) ($v =$) $f\lambda$ or $2(000) \times 16$ C1 $32\,000\text{ cm / s}$ or 320 m / s A1
 10(c)(ii)1 a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest B1 10(c)(ii)2 1.6 cm (if highest frequency is 20 000 Hz) B1
 10(c)(iii) tube or other method to produce (narrow) beam of sound on source and/or on detector B1 stated detector or stated reflector B1 detector moved to find maximum loudness or angles i and r measured with suitable experiment B1 5054/21 Question Answer Marks

Question 6

- 6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b) A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i) Calculate the wavelength of the sound.

wavelength =[2]

- (ii) A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

Mark scheme — 6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1 6(b)(i) ($\lambda =$) v / f or 330 / 3800 B1 0.087 m or 8.7 cm B1 6(b)(ii) not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency B1 Question Answer Marks

2018

May/June 5054/21

Question 11

11 A water wave is produced in a ripple tank.

- (a) (i) Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....[4]

- (ii) The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

- (iii) State what is meant by a *wavefront*.

.....

.....

.....[2]

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

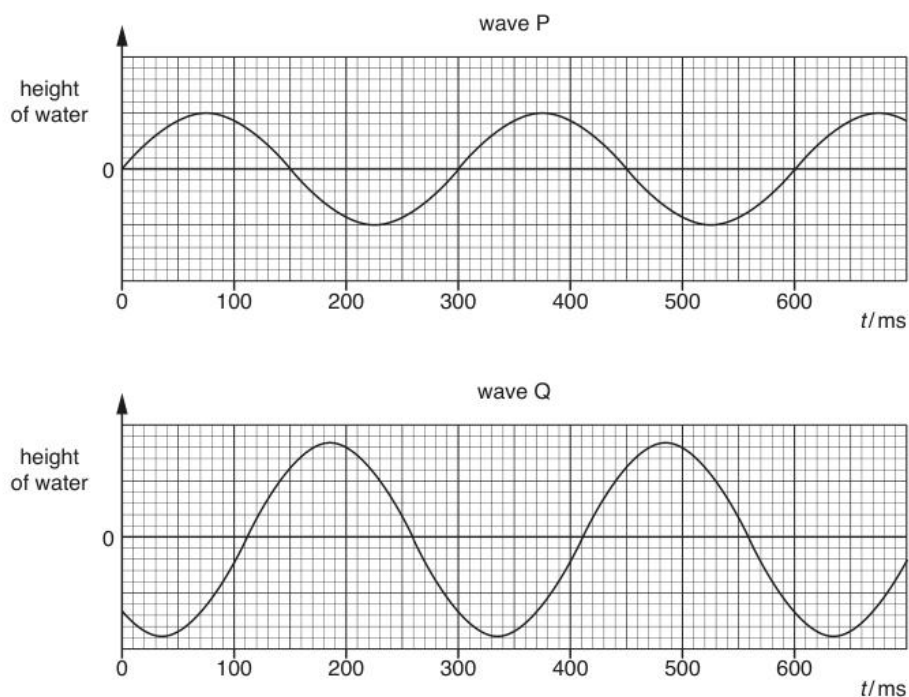


Fig. 11.1

- (i) State one difference and one similarity of the two waves.

difference

.....

similarity

.....

[2]

- (ii) The speed of the water wave is 0.20 m/s.

Determine

1. the frequency of wave P,

frequency =[3]

2. the wavelength of wave P.

wavelength =[2]

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5054/21/M/J/18

Mark scheme — 11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1 11(a)(ii) observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object B1 moves up and down B1 11(a)(iii) line joining points M1 that are at the same point in the motion A1 11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1 11(b)(ii) time 300 (ms) or 0.3 (s) C1 (f=) 1 / t numerical or algebraic in any form C1 3.3 Hz A1 11(b)(ii) 2 ($\lambda =$) v / f or $v = \lambda f$ – numerical or algebraic in any form C1 0.060 m A1

Question 5

2

Section A

Answer **all** the questions in this section. Answer in the spaces provided.

- 1 A student drops a small metal object into a cylinder of oil. The object falls alongside a vertical ruler and a camera records its position at 1.0 s intervals, as shown in Fig. 1.1.

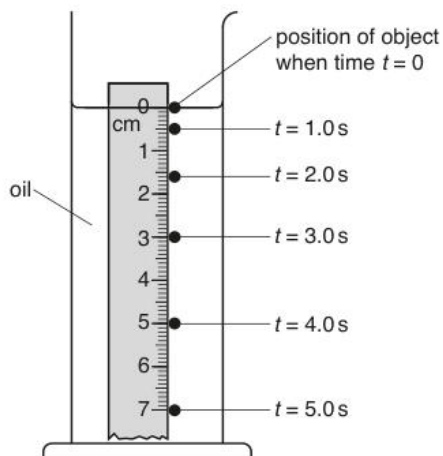


Fig. 1.1

- (a) On the grid in Fig. 1.2, plot a distance-time graph for the object.

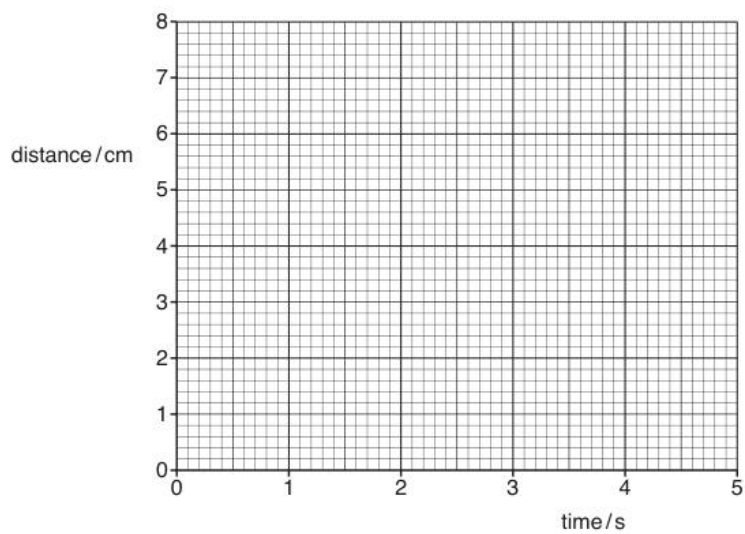


Fig. 1.2

[2]

(b) The object has a constant speed for some of the time.

(i) State and explain how a distance-time graph shows that speed is constant.

.....
.....
.....
.....[2]

(ii) Explain, in terms of the forces acting on the object, how it is able to fall at a constant speed.

.....
.....
.....
.....[2]

- 2 In an experiment, a student takes measurements and determines the extension of a spring for different loads. The apparatus is shown in Fig. 2.1.

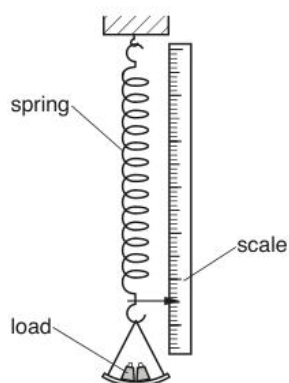


Fig. 2.1

The table in Fig. 2.2 shows the results.

load / N	0	1.0	2.0	3.0	4.0	5.0
length of spring / mm	200	235	270	305	350	420
extension / mm	0	35	70	105		

Fig. 2.2

- (a) Only some of the extensions are shown in the table.

Complete the table to show all of the extensions.

[1]

- (b) Calculate the load that produces an extension of 49 mm.

load =[2]

- (c) The student pulls a load downwards from position A to position B and holds it fixed at position B, as shown in Fig. 2.3.

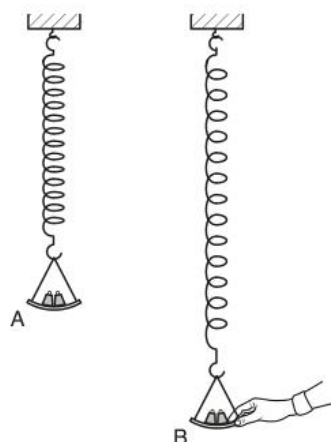


Fig. 2.3

The load is stationary at A and at B. The load has no kinetic energy at either point.

- (i) Place ticks (✓) in the boxes to show how the value of each of the forms of energy compares at A and B.

	equal at A and B	larger at A	larger at B
energy stored in spring	☐	☐	☐
gravitational potential energy of load	☐	☐	☐

[1]

- (ii) Work is done by the student to pull the load down. The law of conservation of energy states that energy cannot be created and cannot be destroyed.

Explain how this principle applies in this case.

.....

 [2]

- 3 Equal volumes of metal, water and air are heated from 20 °C to 80 °C.

(a) State whether the metal, the water or the air expands the most.

.....[1]

- (b) Fig. 3.1 is a diagram of the metal at 20 °C. The metal is a large, square sheet with a square hole at its centre. The sheet lies flat on a table.

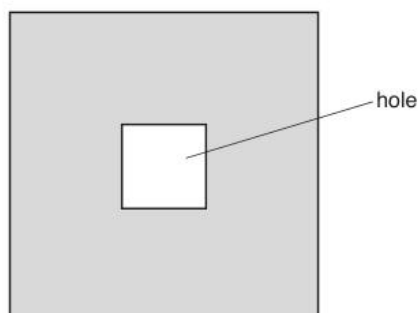


Fig. 3.1

On Fig. 3.1, draw the outline of the sheet and of the hole after the metal is heated. Make sure that the expansion is noticeable. [2]

- (c) The metal has a mass of 5.0 kg and a density of $7.5 \times 10^3 \text{ kg/m}^3$.

(i) Define *density*.

.....
[1]

(ii) Calculate the volume of the metal.

Give your answer to a suitable number of significant figures.

volume = [2]

Mark scheme — 5(a) (d=) $s \times t$ in any form algebraic or numerical, C1 $3.6 \times 10^7 \text{ m}$ A1 5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1 5(b)(ii) Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric) B1 Question Answer Marks

Question 9

- 9 Some waves are longitudinal and some waves are transverse.

(a) State how a longitudinal wave differs from a transverse wave.

.....
 [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

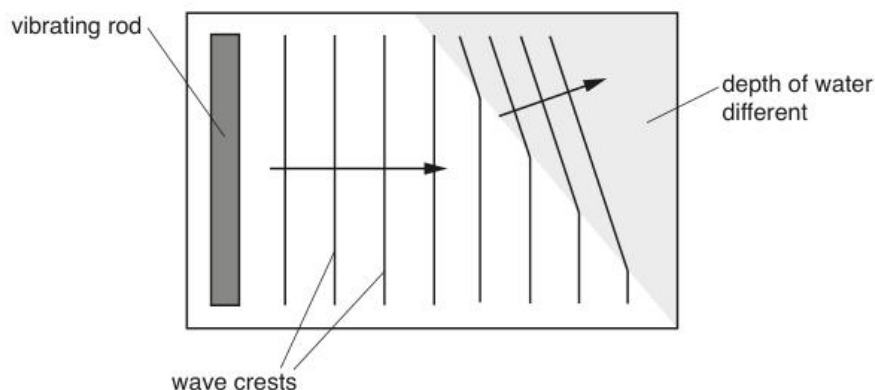


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

- (i) In the left-hand section of the tank, the wavelength of the wave is 0.019 m and it travels at 0.17 m/s.

Calculate the frequency of the wave.

frequency = [2]

- (ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

- (iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

 [2]

- (c) Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60° .

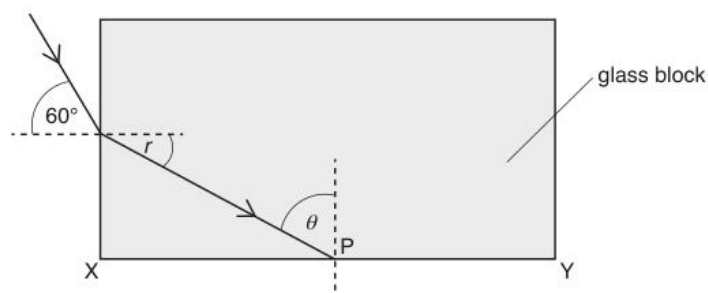


Fig. 9.2

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

- (i) Underline all the terms that describe a light wave.

electromagnetic

longitudinal

transverse

[1]

- (ii) At the point where the light enters the glass, the angle of refraction is r .

Calculate angle r .

$r = \dots\dots\dots$ [2]

- (iii) 1. Calculate the critical angle c for light travelling in the block.

$c = \dots\dots\dots$ [2]

2. At P, the angle θ between the ray and the normal is given by $\theta = 90^\circ - r$.

State and explain what happens to the light when it strikes side XY.

.....

 [2]

3. On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

Mark scheme — 9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1 9(b)(i) ($f =$) v / λ or $0.17 / 0.019$ C1 8.9 Hz A1 9(b)(ii) (frequency) stays constant or does not change B1 9(b)(iii) (speed) decreases B1 (Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave B1 9(c)(i) electromagnetic and transverse underlined B1 9(c)(ii) $n = \sin(i) / \sin(r)$ or ($r =$) $\sin^{-1}(\sin(i) / n)$ or ($r =$) $\sin^{-1}(\sin(60^\circ) / 1.6)$ C1 33° A1 5054/21 Question Answer Marks 9(c)(iii) $1/n = 1 / \sin(c)$ or ($c =$) $\sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$ C1 39° A1 2 total internal reflection B1 $\theta > c$ or $57^\circ > 39^\circ$ or angle of incidence greater than the critical angle B1 3 reflected ray at P (correct angle by eye) B1 light in air at 30° (by eye) to vertical surface B1 Question Answer Marks

Question 9

9 Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....
 [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....

 [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.



Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]
3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

- (ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

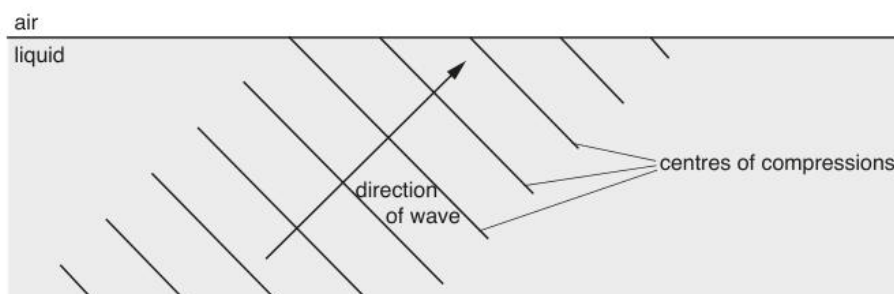


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

- (d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

Mark scheme — 9(a) frequency of sound wave small(er) or its frequency is less than 20 000 Hz B1 9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1 9(b)(ii) 1 two centres of rarefactions labelled R B1 2 distance from one point to adjacent identical point indicated (with double-headed arrow) B1 3. ($v = f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$ C1 1200 m / s A1 9(c)(i) decreases B1 9(c)(ii) four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines B1 at least four compressions in air parallel to each other B1 at least four straight lines at shallower angle from horizontal and slope correct B1 5054/22 Question Answer Marks 9(d) object (to be cleaned) immersed in liquid / solvent B1 object / liquid agitated / vibrated by ultrasound B1 dirt (particles) shaken off or dislodges / removes dirt B1 Question Answer Marks

2019

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Question 10

10 In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

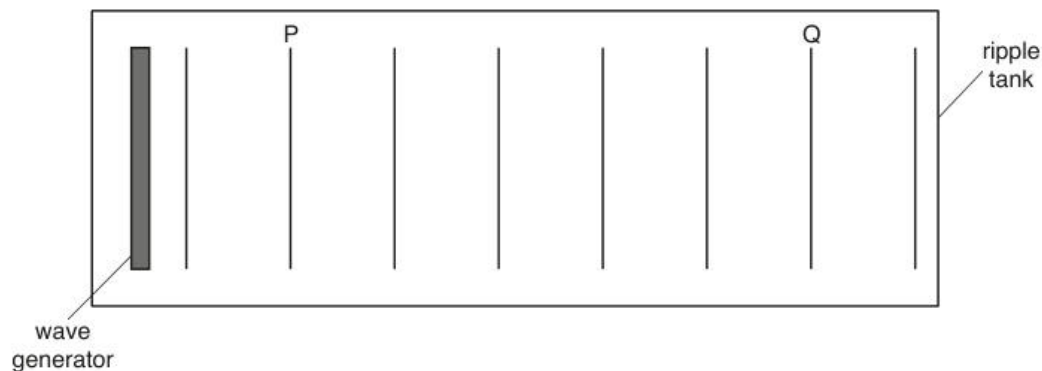


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

(ii) Define the term *frequency*.

.....
 [2]

(iii) Determine the frequency of the wave.

frequency = [1]

- (iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....

 [2]

- (v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed
 wavelength [1]

- (b) A student reads in a textbook that the speed of sound in air is about 300 m/s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

- (i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1.
 2. [2]

- (ii) Describe how all the measurements are made.

.....

 [3]

(iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

[2]

[Total: 15]

Mark scheme — 10(a)(i) 1.7 (cm) seen or any multiple of wavelength measured on diagram C1 6.4–7.2 cm A1 10(a)(ii) number of waves/cycles/oscillations in one second / unit time C1 number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time or number of oscillations of a source/particle in one second / unit time A1 10(a)(iii) 2.5 Hz B1 10(a)(iv) moved up and down B1 regularly or at constant frequency / 2.5 times a second B1 10(a)(v) speed smaller / decreases and wavelength smaller / decreases B1 10(b)(i) tape measure or trundle wheel or microphones B1 stopwatch or timer B1 10(b)(ii) start timer on seeing the smoke from gun / sound picked up by microphone B1 stop timer on hearing sound B1 measure (large) distance between students B1 10(b)(iii) both between and including 1000 and 10 000 m/s B1 solid speed > liquid speed B1 5054/21 Question Answer Marks

Question 5

5 Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

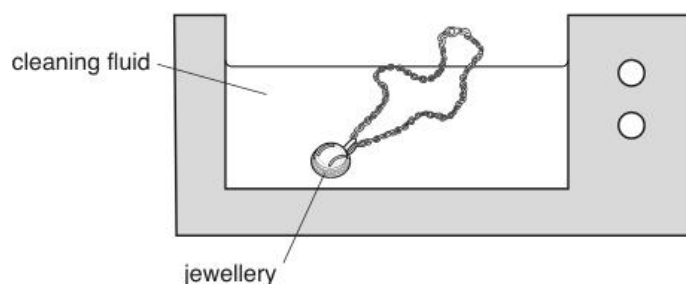


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 Question Answer Marks 5(b)(i) ($\lambda =$) v / f or $1500 / 42\,000$ C1 0.036 m or 3.6 cm A1 5(b)(ii) (ultrasound) vibrates (the cleaning fluid / jewellery) B1 dirt shaken off B1 Question Answer Marks

Question 10

16

- 10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

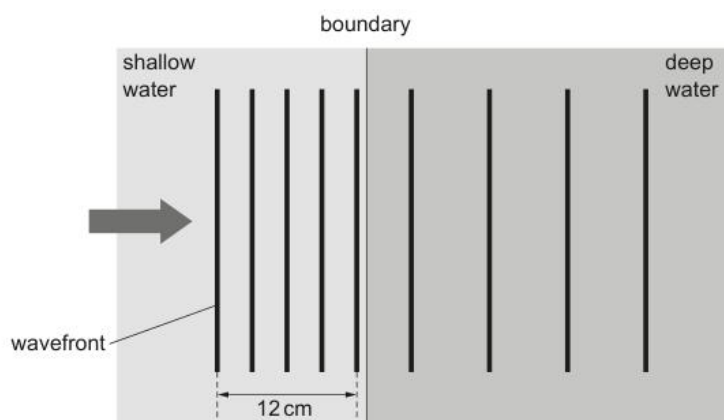


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (i) Determine the wavelength of the wave in the shallow water.

wavelength = [1]

- (ii) Calculate the frequency of the wave produced.

frequency = [2]

- (iii) Determine the speed of the water wave in the shallow water.

speed = [2]

- (iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

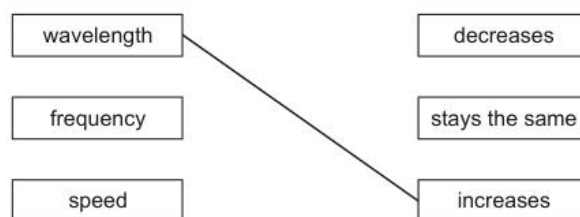


Fig. 10.2

[1]

- (v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

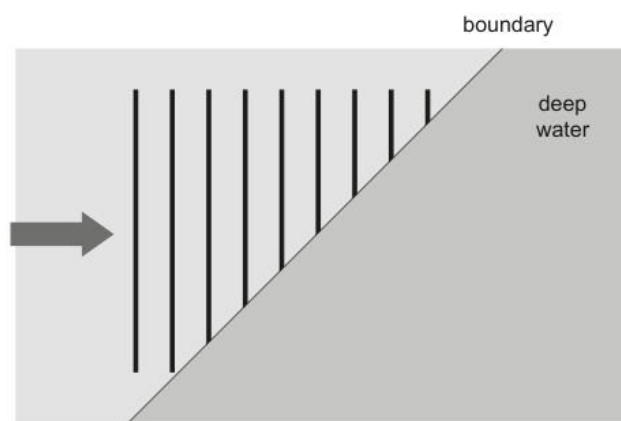


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water.

[3]

(b) White light is made up of different colours.

- (i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength

.....

.....

largest wavelength

[2]

- (ii) A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

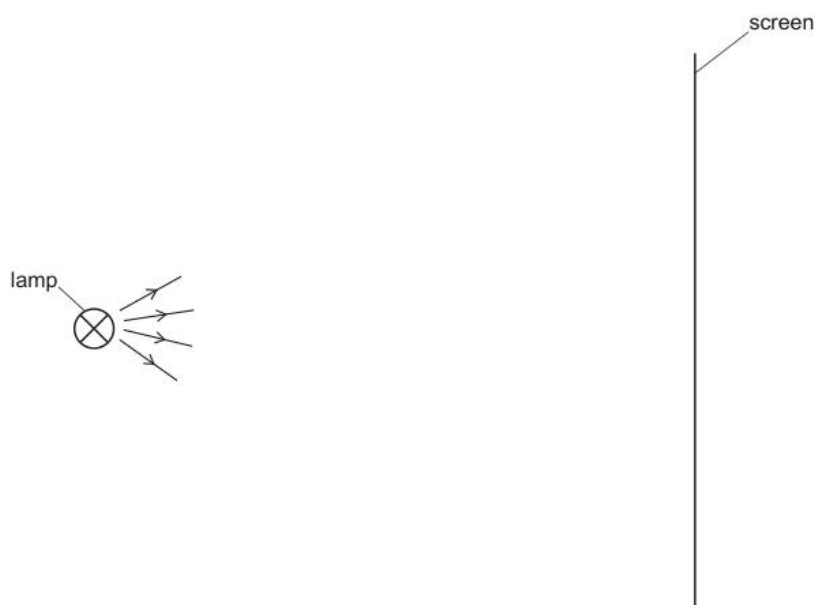


Fig. 10.4

[4]

[Total: 15]

Mark scheme — 10(a)(i) 3 (.0) cm B1 10(a)(ii) number of waves per second or 5 / 10 or $f = 1 / T$ C1 2(.0) Hz A1 10(a)(iii) $(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form C1 6(.0) cm / s A1 10(a)(iv) frequency stays the same and speed increases B1 5054/22 Question Answer Marks 10(a)(v) ■ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between B1 constant wavelength shown between all refracted wavefronts B1 all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal B1 10(b)(i) four colours from violet, indigo, blue, green, yellow, orange, red B1 correct order of four of the above colours B1 10(b)(ii) prism B1 slit (to produce narrow beam) B1 dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face B1 correct refraction on first and second faces for all rays shown B1 Question Answer Marks

2022

May/June 5054/22

Question 8

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

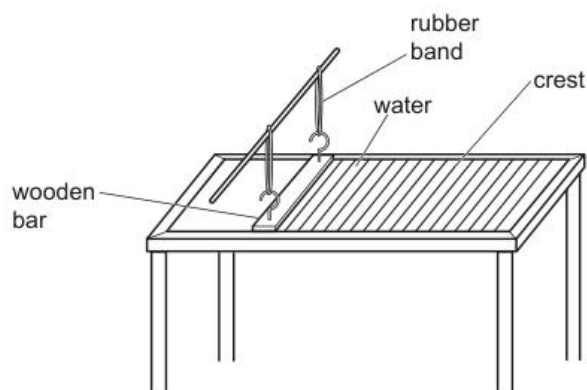


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

- (i) Calculate the number of crests produced in 1.5 s.

number = [1]

- (ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

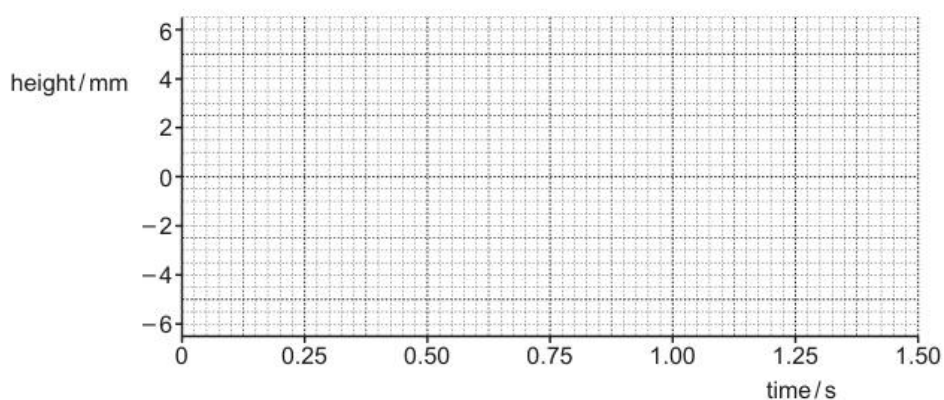


Fig. 8.2

[2]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
 [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed
 wavelength [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

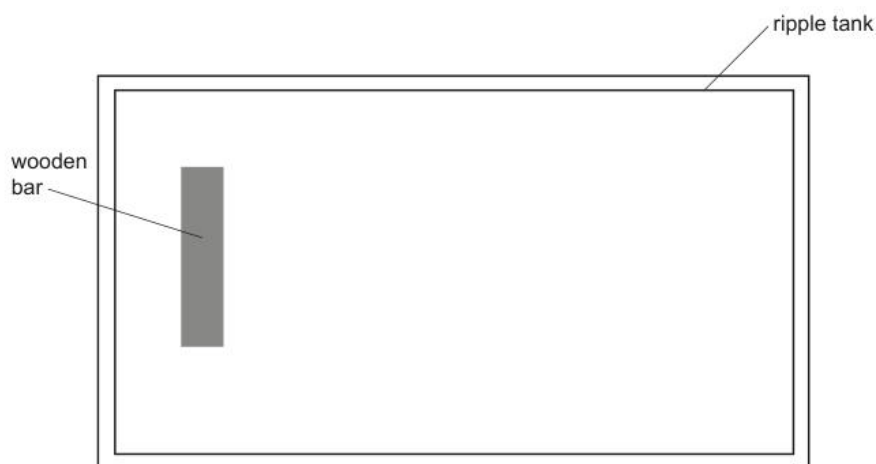


Fig. 8.3

[3]

- (d) Fig. 8.4 shows a connection to the internet made from a remote station A using a satellite above the Earth's surface.

Data is sent between stations A and B using microwaves which travel to and from the satellite.

Station B is connected directly to the internet using optical fibre.

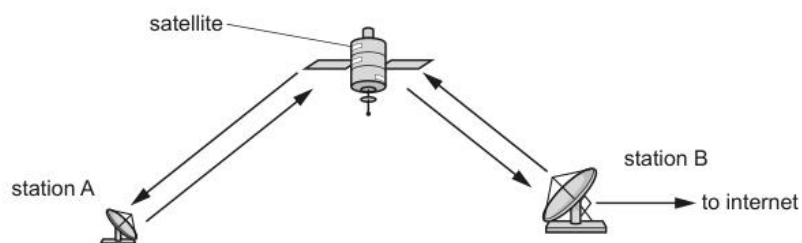


Fig. 8.4 (not to scale)

The speed of microwaves is 3.0×10^8 m/s.

Stations A and B are each 560 km from the satellite.

- (i) Calculate the time taken for data to travel from A to B using microwaves.

time = [3]

- (ii) The same data can be sent from A to B in a shorter time along the surface of the Earth using infrared waves to carry the information in optical fibres.

State **one** property that infrared waves and microwaves have in common.

.....
 [1]

- (iii) Suggest **one** other advantage of using optical fibres to connect stations A and B directly.

.....
 [1]

[Total: 15]

Question 9 begins over the page.

Mark scheme — 8(a)(i) 3 B1 8(a)(ii) shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change) B1 1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i) B1 8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1 8(b)(ii) speed remains the same B1 wavelength decreases B1 8(c)(i) (piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary B1 8(c)(ii) diagram showing within the ripple tank: ■ crests parallel to bar hitting boundary with a boundary labelled B1 ■ correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change B1 ■ different wavelength shown before and after refraction and constant before and after (by eye) B1 8(d)(i) (t =) distance / speed algebraic or numerical seen in any form C1 2 ■ 560(000) seen or used or 0.0019 (s) C1 0.0037 s A1 5054/22 Question Answer Marks 8(d)(ii) One of (both): ■ reflect / refract ■ have same speed (in a vacuum) ■ travel in a vacuum / don't need a medium ■ are electromagnetic ■ are transverse ■ carry energy / cause heating ■ have larger wavelength / smaller frequency than visible light B1

8(d)(iii) less likely to be hacked / intercepted / greater security or more channels / better bandwidth possible / more data or less likely for transmission to be interrupted or cost of satellites saved / satellites expensive B1 Question Answer Marks

Oct/Nov 5054/22

Question 4

4 A sound wave, travelling in air or water, contains compressions and rarefactions.

- (a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

- (b) When a sound wave passes through air or water, the molecules of the air or water move.

Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

- (c) Describe what is meant by 'a rarefaction'.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

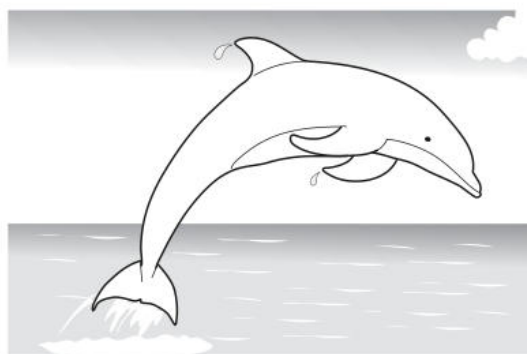


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m/s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = [2]

[Total: 8]

Mark scheme — 4(a) equipment e.g. (evacuatable) container and (suspended electric) bell / source of sound and (air) pump B1 action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced) B1 observation e.g. transmission / hearing of sound ceases / no sound B1 5054/22 Question Answer Marks 4(b) (they) vibrate M1 (vibration) parallel to / in the direction of propagation / of energy travel A1 4(c) (point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average) B1 4(d) (■ =) v / f or $1500 / 140\,000$ C1 1.1 ■ 10–2 m or 1.1 cm or 11 mm A1 Question Answer Marks

Question 7

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

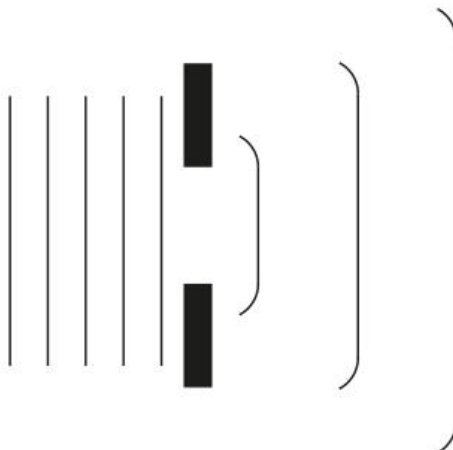


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

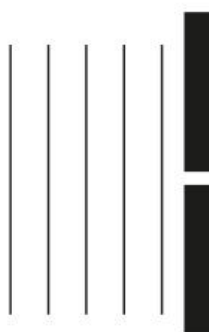


Fig. 7.2

[2]

- (c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm.

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s.

- (i) Define the term 'wavelength'.

.....
 [1]

- (ii) Calculate the frequency of the wave.

frequency = Hz [2]

- (iii) Calculate the wavelength of the wave in the shallow part of the tank.

wavelength = cm [1]

[Total: 7]

Mark scheme — 7(a) wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests B1 7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1 7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1 7(c)(ii) $v = f \lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz) C1 3(0) Hz) A1 7(c)(iii) 1.3 (cm) B1 5054/21 Question Answer Marks

Question 5

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves seismic S-waves sound X-rays [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

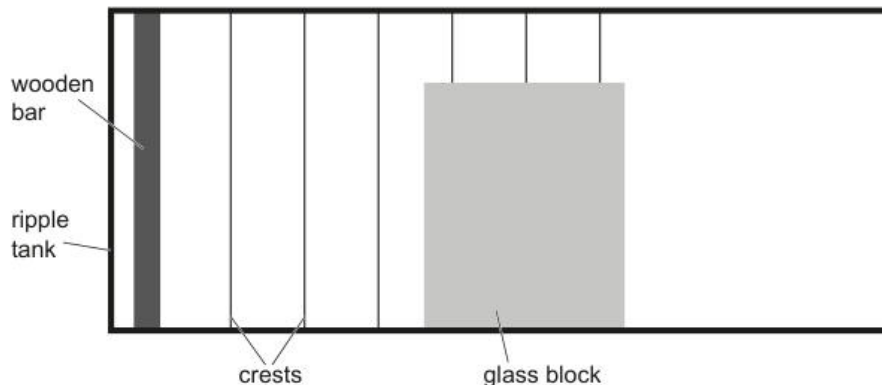


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

Mark scheme — 5(a) seismic S-waves and X-rays B1 5(b)(i) (wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$ C1 frequency 6.7 (Hz) A1 wavelength 4.0 (cm) A1 5(b)(ii) straight wavefronts above block to the right with same wavelength B1 correct diffraction – wavefronts quarter circles with centre at top corner B1 5(b)(iii) increase frequency of movement of (wooden) bar B1 5054/22 Question Answer Marks 5(b)(iv) less diffraction or less bending B1 Question Answer Marks

Question 4

2

- 1 A ball is released from rest at point A and moves along a smooth track ABCDE as shown in Fig. 1.1. The ball is shown at point A and as it passes point B.

The ball is always in contact with the track and air resistance is negligible.

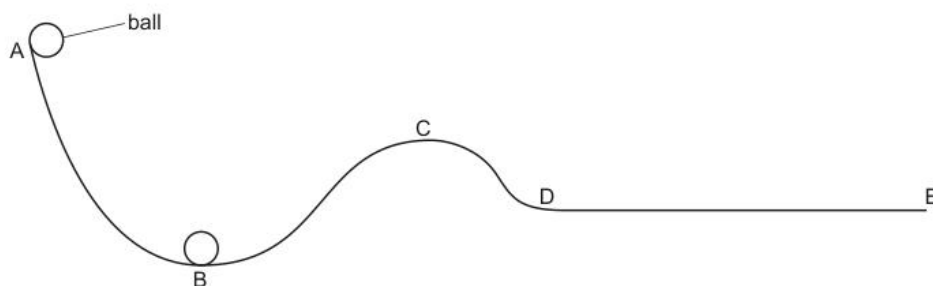


Fig. 1.1

Fig. 1.2 shows the distance–time graph for the ball as it moves from A to E.

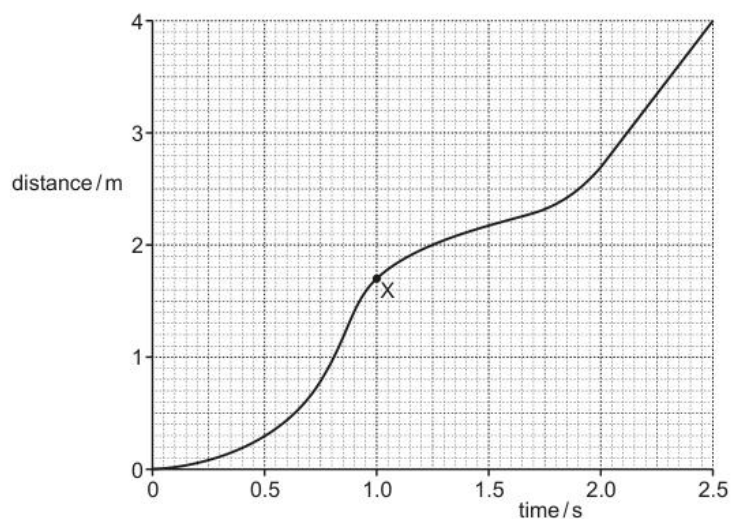


Fig. 1.2

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

- (ii) Determine the speed of the ball at point X on Fig. 1.2.

Show your working.

speed = m/s [3]

- (b) The speeds of the ball at A, B, C and D are v_A , v_B , v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest —————→ fastest

--	--	--	--

[1]

- (c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

- (d) The track at point B is circular in shape.

On Fig. 1.1, draw an arrow to show the direction of the resultant force on the ball at point B.

[1]

[Total: 10]

- 2 A small aircraft takes off from the horizontal deck of a ship.

Before taking off, the aircraft is held in place by a holdback bar.

When the holdback bar is released, the aircraft is pulled along the deck by a steam-powered piston as shown in Fig. 2.1.

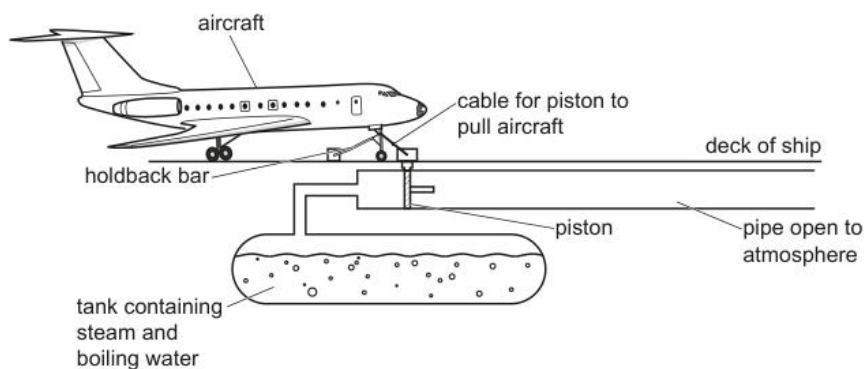


Fig. 2.1 (not to scale)

The steam exerts a high pressure on the piston.

- (a) (i) Explain, using ideas about particles, how the steam creates a pressure on the piston.

.....

.....

.....

..... [2]

- (ii) Explain why the pressure on the piston increases as the temperature of the steam increases.

.....

.....

.....

..... [2]

- (b) When the pressure is high enough, the holdback bar is released. The steam pushes the piston along the pipe shown in Fig. 2.1.

The piston has a cross-sectional area of 0.30 m^2 .

The pressure of the steam in the tank is $2.1 \times 10^6 \text{ Pa}$.

Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

Determine the resultant force on the piston caused by the pressure difference.

Show your working.

force = N [2]

- (c) The force calculated in (b) causes the aircraft to accelerate to a maximum speed of 28 m/s from rest.

The mass of the aircraft is $3.0 \times 10^4 \text{ kg}$.

- (i) Calculate the momentum of the aircraft at the maximum speed.

momentum = kg m/s [2]

- (ii) Calculate the time taken for the aircraft to reach the maximum speed.

time = s [2]

[Total: 10]

- 3 A pan containing ice at -15°C is placed on a gas heater as shown in Fig. 3.1.



Fig. 3.1

- (a) Thermal conduction occurs in the metal from which the pan is made.

By referring to particles, describe the process of thermal conduction in a metal.

.....

.....

..... [2]

- (b) Fig. 3.2 shows how the temperature of the contents of the pan varies with time.

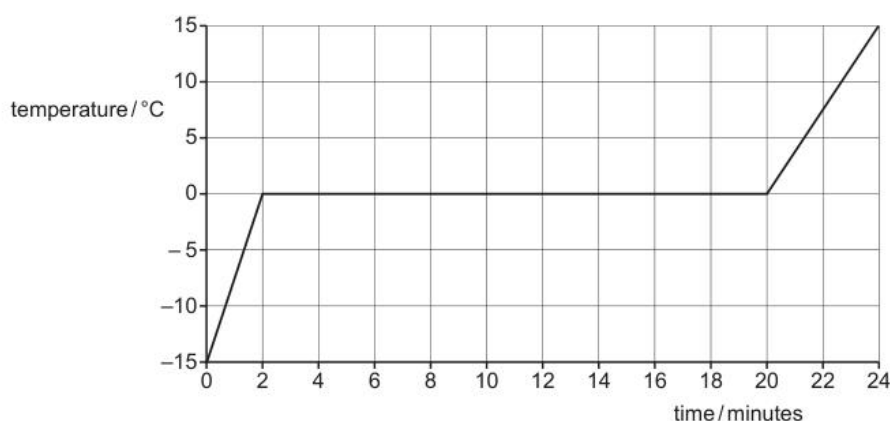


Fig. 3.2

Mark scheme — 4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) ($V =$) $f\lambda$ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1 4(b)(ii) earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning B1 4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

Question 5

- 5 Ultrasound is a longitudinal wave which cannot be heard by humans as the frequency of ultrasound is too high.

- (a) Ultrasound waves **cannot** travel in a vacuum.

Explain why.

.....

 [2]

- (b) Describe how a longitudinal wave differs from a transverse wave.

.....

 [2]

- (c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

- (i) Calculate the wavelength of this ultrasound wave in soft human tissue.

wavelength = m [3]

- (ii) The ultrasound wave passes from the soft tissue into bone where the speed of the ultrasound wave is greater than 1500 m/s.

State what happens to the frequency and to the wavelength of the ultrasound wave as it passes into the bone.

frequency
 wavelength [2]

[Total: 9]



Mark scheme — 5(a) ultrasound / sound / it is transmitted by vibrating matter / particles B1 no matter / particles in a vacuum / no medium B1 5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1 5(c)(i) (■ =) v / f or $1500 / 8.4$ ■ 106 C1 1.8 ■ 10N C1 1.8 ■ 10–4 (m) A1 5(c)(ii) frequency: does not change B1 wavelength: increases B1 5054/21 Question Answer Marks

Question 5

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
-
-
- [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.
- State what happens to the speed and what happens to the wavelength of the wave produced.
- speed
- wavelength [2]





(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

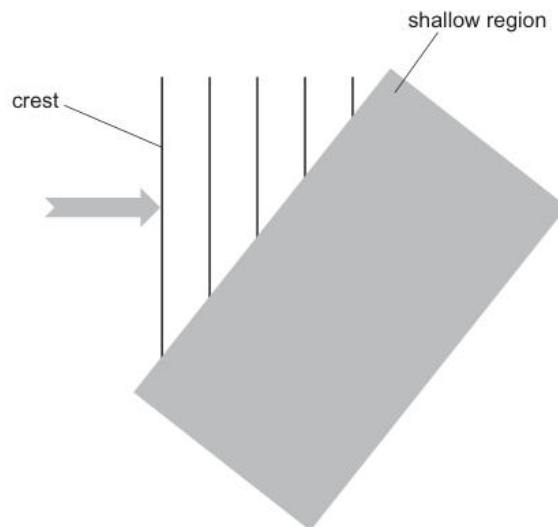


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



Mark scheme — 5(a)(i) C and R marked correctly B1 5(a)(ii) compressions are where pressure is high / particles close together / density is high B1 5(b)(i) 0.75 (Hz) B1 5(b)(ii) speed: no change B1 wavelength: decreases B1 5(b)(iii) at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal M1 refracted towards the surface on correct side of surface and join wavefronts shown A1 5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 Question Answer Marks

Question 4



6

- 4 A sound wave from a vibrating tuning fork is displayed on the oscilloscope shown in Fig. 4.1. The sound wave is detected using a microphone.

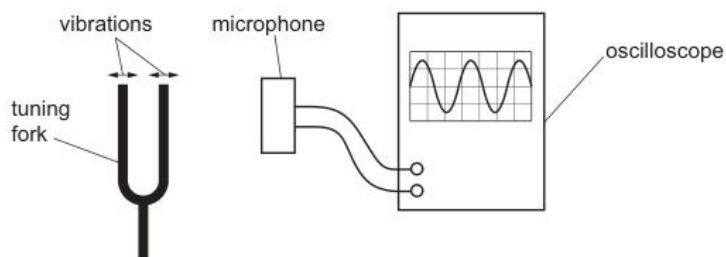


Fig. 4.1

- (a) Describe the motion of the air particles as the sound wave passes from the tuning fork to the microphone.

.....

 [2]

- (b) Fig. 4.2 shows the trace seen on the oscilloscope when the tuning fork vibrates.

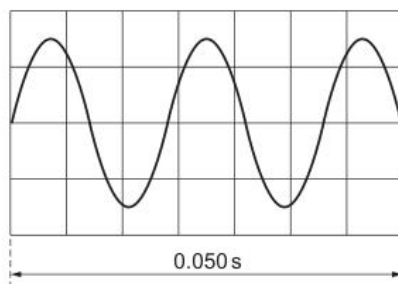


Fig. 4.2

The frequency of the sound is constant.

- (i) State what is meant by 'frequency'.

.....
 [1]

- (ii) The trace shown on Fig. 4.2 is produced in 0.050 s.

Calculate the frequency of the sound wave.





7

- (iii) A student near the tuning fork observes the trace and listens to the sound. He notices that the amplitude of the trace decreases.

State how the sound heard by the student changes.

..... [1]

- (iv) The tuning fork is replaced by a new tuning fork that produces a sound wave of half the frequency. The controls on the oscilloscope are not changed.

On Fig. 4.3, draw the trace obtained with the new tuning fork. The trace obtained with the original tuning fork is shown.

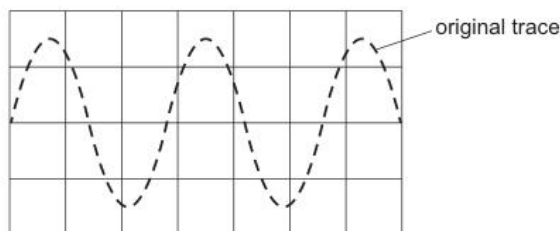


Fig. 4.3

[1]

- (c) Fig. 4.4 shows three adjacent wavefronts of the sound wave as the wavefronts move towards a gap in a wall created by an open door.

Another student, standing at P on the other side of the wall, hears the sound.

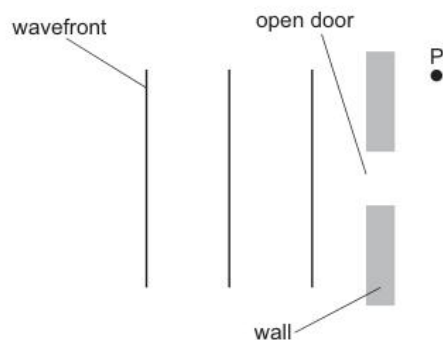


Fig. 4.4

- (i) On Fig. 4.4, draw the three wavefronts after they pass through the gap. [2]

- (ii) State the name of the process involved as the wavefronts pass through the gap and reach P.

..... [1]

[Total: 10]

